- *Application-level programming* --This level involves programming using web (or enterprise or mobile) application frameworks and user interface design concepts that are outside the blockchain protocol. In chapters 5-11, you'll be provided details on web programming to link to underlying smart contracts and to deploy end-to-end *decentralized applications (Dapps)* on the blockchain.

> **DEFINITION** *Dapps* are web or enterprise applications that include application logic to invoke blockchain functions that implement trust intermediation.

Dapps embed a significant code element--that of smart contracts. For any given smart contract, an exact copy of the smart contract's code is transmitted through a special transaction and deployed in the participant nodes of a blockchain network.

> **DEFINITION** A *smart contract* is an immutable executable code representing the logic of a Dapp. The data variables and functions defined in a smart contract collectively represent the state and operations for enforcing an application's (Dapp's) rules for verification, validation, and recording on the blockchain.

1.3 Blockchain programming

In the evolution from sequential programming to structured programming, functional programming, object-oriented programming (OOP), web and database programming, and big data programming, programmers experienced shifts in approaches, artifacts, and architectures (such as OOP with classes and objects, and Hadoop and Map Reduce for big